Ashim Paudel

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EDUCATION

MS in Computer Science | Louisiana State University | Expected Dec 2026
GPA: 4.0 | Relevant Courses: Machine Learning, Deep Learning, Applied LLM Development, Computational Chemistry

BS in Computer Science | Mississippi State University | May 2022
GPA: 3.41 | Relevant Courses: DBMS, Programming Languages, AI Robotics, Algorithms, Microprocessors

TECHNICAL SKILLS

Programming Languages: Python, C, C++, Java, MATLAB, Bash
Frameworks/Libraries: PyTorch, TensorFlow, Scikit-Learn, Transformers, OpenCV, NumPy, Pandas
Agentic Workflows: LangChain, LangGraph, MCP, RAG, FastAPI
Database/Technologies: MySQL, SparkSQL, Apache Kafka, Apache Hadoop, Git, Docker
AI Tools: Claude Code, Cursor, Codex, Copilot
Scientific Computing: NAMD, GROMACS, VMD, HPC, SLURM

EXPERIENCE

Graduate Research Assistant Jan 2025 - Present
Louisiana State University | Baton Rouge, LA

- Architected a graph neural network (GNN) active learning framework using MARTINI coarse-grained molecular graphs and query-by-committee sampling to predict polymer–drug encapsulation, requiring 45% fewer GROMACS simulations than uniform random sampling.
- Developed AI agents to automate molecular dynamics workflows, including molecular structure parsing, simulation setup, execution, and trajectory analysis.
- Developed an LLM-assisted scientific literature extraction pipeline using prompt engineering to extract and structure polymer and molecular simulation data from research publications.
- Applied spaCy NLP to extract polymer and force-field mentions from scientific papers, detecting $\sim 8\times$ more polymer name variants than regex parsing.
- Performed molecular dynamics simulations of polymer–drug systems, including topology generation, system solvation, and trajectory analysis; used Bash scripts to automate multi-step workflows and Slurm for HPC job execution.

Software Engineer Jul 2022 - Jan 2024
Orbbec | Troy, MI

- Developed a real-time image processing pipeline in C++ for converting RGB-D images into a stereoscopic view. Integrated Orbbec's proprietary algorithm, leveraging OpenCV and Boost for efficient multithreaded data handling, and used StereoKit for XR rendering across VR platforms.
- Developed "The Eye for AIs," an advanced image processing application that isolates objects, computes volumes, and performs detailed analyses using color and depth images from a depth camera. Leveraged vision-language models (VLMs) and large language models (LLMs) to segment and interpret regions of interest in images.
- Led the integration of a tailored LLM API into a customer chatbot application. Improved technical response accuracy by 18% by fine-tuning the model using the company's software development kit, product specifications, and API documentation.
- Led the design of an iOS prototype to capture and transmit synchronized RGB camera and LiDAR depth sensor data over a network for real-time stereoscopic 3D reconstruction. Conducted in-depth research on Apple's LiDAR sensor and depth APIs.
- Collaborated with NVIDIA to evaluate the compatibility of its body- and hand-tracking technologies with the Orin and Jetson Nano platforms. Conducted thorough testing of Orbbec SDKs, documenting test cases, outcomes, and errors.

PUBLICATIONS

- Paudel, A., Shakya, A., et al. A structured dataset of polymers in molecular dynamics simulations extracted by LLM-assisted text mining. Scientific Data (2026). Accepted.